

August 31, 2026

EIM

Dear Hiring Team,

I am writing to apply for the Data Analyst & AI Engineer position at EIM. I recently completed my M.Sc. in Computer Science at the University of Calgary, where my work focused on Python/C++ data pipelines, machine learning, structured data systems, retrieval workflows, and evaluation for large scientific datasets. I am especially interested in EIM because the role applies AI to real operational drilling data rather than abstract demos, which is the kind of practical data problem I enjoy.

My graduate research involved building software for large Earth observation datasets. I developed Python and C++ pipelines for data collection, preprocessing, validation, storage, retrieval, analysis, and model evaluation. One major project was a DGGS-based framework for managing and retrieving large volumes of spatiotemporal satellite data. I designed multiresolution storage and retrieval methods, implemented performance-sensitive C++ components, and evaluated the system using large Canada-wide datasets. This experience maps well to building data analysis pipelines where structure, queryability, and reliability matter.

I also worked on a digital elevation model super-resolution project that strengthened my experience with machine learning validation and correction-style workflows. I built Python pipelines using NumPy, pandas, Rasterio, Google Earth Engine, and PyTorch; trained deep learning models; and evaluated outputs with RMSE, slope, TRI, TPI, and wavelet-based metrics. A key part of the work was identifying when an evaluation setup made results look better than they really were, then designing stricter validation methods to make the conclusions more useful.

In addition to my research background, I bring backend experience from Asr Gooyesh Pardaz, where I developed Django REST APIs backed by PostgreSQL for NLP, text-to-speech, and speech-to-text products. I worked with authentication features, payment-service integration, relational records, and a WebRTC chatbot system using GPT-style responses and FAQ data. I have also been building an agentic job-search automation platform using LLM workflows, Google Sheets, Google Drive, Telegram API, Docker, and workflow automation tools, which has given me practical exposure to AI-assisted search, matching, and document-oriented workflows.

EIM's focus on AI-powered search, document interaction, structured and semi-structured drilling datasets, vector retrieval, and ML-based validation is a strong fit with my background in data pipelines, retrieval-oriented systems, and model evaluation. I have less direct drilling-domain experience, but I am based in Calgary, comfortable learning domain-specific data through careful analysis and collaboration, and motivated by the chance to build tools used in real engineering and operational environments.

Thank you for considering my application. I would welcome the opportunity to discuss how my Python, SQL, machine-learning, retrieval, and data-pipeline experience could contribute to EIM's drilling intelligence platform.

Sincerely,

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